EcoPick - Development Roadmap

Tổng quan Timeline

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Phase 1 (MVP)          Phase 2 (Social)       Phase 3 (Growth)       Phase 4
(Scale)
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8 tuần                 6 tuần                 6 tuần                 Ongoing

Auth + Map + Report    Social Feed + Chat     Marketplace + Tour     AI +
Analytics
Events + Profile       Leaderboard + Rewards  Advanced Gamification  Scale &
Optimize
Notifications          Follow System          Admin Dashboard
Monetization
```

Phase 1: MVP Core (8 tuần)

> **Mục tiêu**: App hoạt động end-to-end — người dùng đăng ký, báo cáo rác, tham gia sự kiện, nhận điểm.

Sprint 1-2: Foundation (Tuần 1-2)

Backend

Task	Priority	Effort
Setup NestJS project, PostgreSQL, Redis	P0	2d
Database migration (users, reports, events tables)	P0	2d
Auth module: register, login, JWT, refresh token	P0	3d
Google OAuth integration	P0	1d
Apple Sign-In integration	P0	1d
Email verification (OTP)	P1	1d
Forgot/reset password flow	P1	1d
Upload module: S3 presigned URLs	P0	1d
Image processing worker (resize, thumbnail)	P1	1d

Frontend

Task	Priority	Effort
Auth screens: Login, Register, Forgot Password, Verify Email	P0	3d

Task	Priority	Effort
Auth state management (SecureStore for tokens)	P0	1d
Auto-login flow (check stored refresh token on launch)	P0	1d
Protected routes (redirect to login if not authenticated)	P0	1d
Splash screen → auth check → login or main tabs	P0	0.5d
API client setup (Axios/fetch with interceptors, auto-refresh)	P0	1d

Deliverables

- ☐ User có thể đăng ký, đăng nhập, đăng xuất
- ☐ JWT auth hoạt động end-to-end
- ☐ Upload ảnh lên S3 thành công
- ☐ Protected API routes verify token

Sprint 3-4: Map & Report (Tuần 3-4)

Backend

Task	Priority	Effort
Reports CRUD API	P0	2d
PostGIS nearby query (GET /reports/nearby)	P0	2d
Reverse geocoding (lat/lng → address)	P1	1d
Report photo management	P0	1d
Upvote/downvote report API	P1	0.5d
Report status workflow (reported → verified → cleaned)	P1	1d
Comments API (for reports)	P1	1d
Points system: earn points on report creation	P0	1d
Badge check worker (trigger after points change)	P1	1d

Frontend

Task	Priority	Effort
Connect Map screen to real API (replace mock data)	P0	2d
Report form → API upload flow (presigned URL → upload → create report)	P0	2d
Location Detail screen → real data	P0	1d
Upvote functionality	P1	0.5d

Task	Priority	Effort
Comments section → real API	P1	1d
Offline report draft (queue for later sync)	P2	1.5d
Error handling & loading states cho map/report	P0	1d
Pull-to-refresh cho map markers	P1	0.5d

Deliverables

- ☐ Bản đồ hiển thị reports thật từ database
- ☐ Chụp ảnh → upload → tạo report thành công
- ☐ Nearby query trả về reports trong bán kính
- ☐ Upvote, comment hoạt động
- ☐ Points tự động cộng khi tạo report

Sprint 5-6: Events (Tuần 5-6)

Backend

Task	Priority	Effort
Events CRUD API	P0	2d
Event participants API (join, leave, list)	P0	1.5d
Event nearby query (PostGIS)	P0	1d
Event check-in API (GPS verification)	P1	1d
Points: earn on event join + completion	P0	0.5d
Event status auto-update (cron: upcoming → ongoing → completed)	P1	1d
Event cover image upload	P0	0.5d

Frontend

Task	Priority	Effort
Events list → real API	P0	1.5d
Event Detail → real data	P0	1d
Join/Leave event flow	P0	1d
Create event screen (for organizers)	P1	2d
Event check-in (GPS proximity check)	P1	1d
Event filter & search	P1	1d

Deliverables

- ☐ Danh sách events từ API, filter theo status
- ☐ Join/leave event hoạt động
- ☐ Check-in tại event (GPS verify)
- ☐ Organizer có thể tạo event mới

Sprint 7-8: Profile, Badges & Notifications (Tuần 7-8)

Backend

Task	Priority	Effort
User profile API (get, update, avatar upload)	P0	1.5d
User stats aggregation	P0	1d
Activity log API	P1	1d
Badges system: define all badges + unlock logic	P0	2d
Push notification infrastructure (FCM setup)	P0	2d
Notification API (list, read, unread count)	P0	1d
Notification triggers (report verified, badge earned, event reminder)	P1	2d

Frontend

Task	Priority	Effort
Profile screen → real API	P0	1d
Edit profile screen	P0	1d
Badges screen → real data (earned/locked)	P0	1.5d
Push notification setup (Expo Notifications)	P0	1d
Notification center screen (bell icon + list)	P0	1.5d
Deep linking from notifications	P1	1d
Settings screen (notification preferences)	P1	1d

Deliverables

- ☐ Profile hiển thị dữ liệu thật, edit hoạt động
- ☐ Badges unlock khi đạt milestone
- ☐ Push notifications gửi đúng trigger
- ☐ Notification center hiển thị + đánh dấu đã đọc

Phase 1 — Tổng kết MVP

Acceptance Criteria

- ☐ Auth: đăng ký/đăng nhập/đăng xuất/reset password
- ☐ Map: hiển thị reports thật, filter, nearby, directions
- ☐ Report: chụp ảnh → upload → tạo report → nhận điểm
- ☐ Events: browse, join, check-in, tạo (organizer)
- ☐ Profile: xem/sửa profile, badges, activity log
- ☐ Notifications: push + in-app notification center
- ☐ Points: tự động cộng/trừ cho mọi hoạt động
- ☐ Offline: report draft lưu local
- ☐ Performance: API < 500ms, app load < 2s

Tech Debt cần clear

- ☐ Unit tests cho backend (>80% coverage cho auth, reports)
- ☐ E2E tests cho critical flows (register → login → report → submit)
- ☐ API documentation (Swagger/OpenAPI)
- ☐ Error monitoring (Sentry)
- ☐ CI/CD pipeline

Phase 2: Social & Engagement (6 tuần)

> **Mục tiêu**: Biến app thành mạng xã hội xanh — follow, feed, chat, bảng xếp hạng.

Sprint 9-10: Social Feed (Tuần 9-10)

Backend

Task	Priority	Effort
Posts CRUD API	P0	2d
Post media (multi-image, video)	P0	1.5d
Like/unlike API	P0	0.5d
Comments on posts API	P0	1d
Follow/unfollow API + follower/following count	P0	1.5d
Feed algorithm: Following feed (chronological)	P0	1.5d
Feed algorithm: Discover feed (trending by likes/recency)	P1	2d
Hashtag indexing + trending hashtags	P1	1d
Auto-post: when user creates report or joins event	P2	1d

Frontend

Task	Priority	Effort
Feed screen (Following + Discover tabs)	P0	3d
Post card component (ảnh carousel, like, comment, share)	P0	2d
Create post screen	P0	2d
Like animation (heart burst)	P1	0.5d
Comments bottom sheet	P0	1d
User profile → follow/unfollow button	P0	0.5d
Follower/following list screens	P1	1d
Hashtag search + trending	P1	1d

Deliverables

- ☐ Feed hiển thị bài từ người follow
- ☐ Discover tab hiển thị trending posts
- ☐ Create post với ảnh/video + caption
- ☐ Like, comment, share hoạt động
- ☐ Follow/unfollow, follower/following list

Sprint 11-12: Chat & Leaderboard (Tuần 11-12)

Backend

Task	Priority	Effort
WebSocket server (Socket.IO)	P0	2d
Conversations API (list, create DM, get messages)	P0	2d
Auto-create group chat khi event created	P1	1d
Message persistence + pagination	P0	1d
Read receipts + typing indicators	P2	1d
Leaderboard API (Redis sorted set)	P1	1.5d
Leaderboard: weekly/monthly/all-time + by city	P1	1d

Frontend

Task	Priority	Effort
Conversations list screen	P0	1.5d
Chat room screen (messages, input, send photo)	P0	3d

Task	Priority	Effort
Event group chat integration	P1	1d
Typing indicator, read receipts UI	P2	1d
Leaderboard screen (tabs, rank list, highlight current user)	P1	2d

Deliverables

- ☐ Chat 1-1 hoạt động real-time
- ☐ Group chat cho mỗi event
- ☐ Leaderboard weekly/monthly/all-time
- ☐ Rank animation khi lên/xuống hạng

Sprint 13-14: Reward Store & Polish (Tuần 13-14)

Backend

Task	Priority	Effort
Rewards catalog API	P1	1d
Redeem reward API (deduct points, create order)	P1	1.5d
Redemption history API	P1	0.5d
Daily challenges system	P2	2d
Login streak tracking	P2	1d

Frontend

Task	Priority	Effort
Reward store screen (catalog grid, points balance)	P1	2d
Redeem flow (confirm dialog, animation, success)	P1	1.5d
Redemption history	P1	1d
Daily challenges card (on feed or profile)	P2	1.5d
Polish: loading skeletons, error states, empty states cho all screens	P0	2d
Polish: animations, transitions, haptics	P1	1.5d

Deliverables

- ☐ Reward store với catalog thật
- ☐ Đổi điểm lấy reward thành công
- ☐ Daily challenges hiển thị + hoàn thành
- ☐ Toàn app polish (skeleton, error, empty states)

Phase 3: Growth Features (6 tuần)

> **Mục tiêu**: Marketplace trao đổi đồ, tour du lịch xanh, AI features, admin dashboard.

Sprint 15-16: Marketplace / Exchange (Tuần 15-16)

Backend

Task	Priority	Effort
Exchange items CRUD API	P1	2d
Nearby items (PostGIS)	P1	1d
Request/approve/reject flow	P1	2d
Transaction completion + points	P1	1d
Rating system after transaction	P2	1.5d

Frontend

Task	Priority	Effort
Exchange browse screen (grid, filter, search)	P1	2.5d
Item detail screen	P1	1.5d
Create listing screen (photos, details, category)	P1	2d
Request flow UI (send, manage incoming requests)	P1	2d
Rating dialog after transaction	P2	1d

Deliverables

- [ ] Đăng đồ trao đổi/tặng thành công
- [ ] Browse items theo category, distance
- [ ] Request flow hoạt động end-to-end
- [ ] Rating sau giao dịch

Sprint 17-18: Tour System (Tuần 17-18)

Backend

Task	Priority	Effort
Tour stops CRUD (sub-resource of events)	P1	1.5d
Stop check-in API (GPS proximity 100m)	P1	1.5d
Tour progress tracking	P1	1d

Task	Priority	Effort
Tour completion logic + bonus badge	P1	1d
Route calculation between stops	P2	1d

Frontend

Task	Priority	Effort
Tour detail screen (map + stops)	P1	3d
Stop check-in UI (proximity indicator, check-in button)	P1	2d
Tour progress bar / stepper	P1	1d
Create tour screen (for organizers) — add stops on map	P1	3d

Deliverables

- ☐ Tour hiển thị lộ trình + stops trên map
- ☐ Check-in tại mỗi stop (GPS verify)
- ☐ Hoàn thành tour → bonus badge + points
- ☐ Organizer tạo tour với nhiều stops

Sprint 19-20: AI & Admin (Tuần 19-20)

Backend

Task	Priority	Effort
AI waste classification (TensorFlow Serving / Google Vision API)	P2	3d
Admin API: user management, content moderation	P1	2d
Analytics API: impact dashboard data	P1	2d
Report export (PDF/CSV)	P2	1.5d
Content moderation: auto-flag inappropriate images	P2	2d

Frontend

Task	Priority	Effort
AI auto-detect: khi chụp ảnh → suggest waste types	P2	2d
Admin web dashboard (React) — user management	P1	3d
Admin dashboard — reports moderation	P1	2d
Admin dashboard — analytics charts	P1	2d

Task	Priority	Effort
Impact report screen (personal stats visualization)	P2	2d

Deliverables

- [ ] AI suggest loại rác từ ảnh (>80% accuracy)
- [ ] Admin dashboard quản lý users, reports, events
- [ ] Impact analytics visualization

Phase 4: Scale & Optimize (Ongoing)

> **Mục tiêu**: Tối ưu hiệu năng, mở rộng quy mô, thêm tính năng nâng cao.

Optimization

Task	Priority
Database query optimization (explain analyze, indexes)	P0
API caching strategy (Redis, CDN)	P0
Image optimization pipeline (WebP, lazy loading)	P1
App bundle size optimization (code splitting)	P1
Offline-first architecture (local DB sync)	P2

Scale

Task	Priority
Horizontal scaling (load balancer, multiple instances)	P0
Database read replicas	P1
CDN for all static assets	P0
Monitoring & alerting (Grafana, Prometheus)	P0
Automated backups & disaster recovery	P0

Advanced Features

Task	Priority
Multi-language support (i18n: Vietnamese, English)	P1
Dark mode	P2
Widget (iOS/Android) — nearby reports count	P2
Apple Watch / WearOS companion app	P3

Task	Priority
AR mode — point camera at trash → info overlay	P3
Carbon footprint calculator	P2
Integration with government waste management APIs	P2
Sponsor/partner portal	P2

Tổng kết Resource Estimation

Team Composition đề xuất

Role	Số người	Phase
Backend Developer (Node.js/NestJS)	1-2	All
Mobile Developer (React Native)	1-2	All
UI/UX Designer	1	Phase 1-2
DevOps / Infra	0.5 (part-time)	All
QA / Tester	0.5 (part-time)	Phase 1+
Product Manager	1	All

Timeline tổng

Phase	Duration	Cumulative
Phase 1 — MVP	8 tuần	8 tuần
Phase 2 — Social	6 tuần	14 tuần
Phase 3 — Growth	6 tuần	20 tuần
Phase 4 — Scale	Ongoing	—

Cost Estimation (Infrastructure — Phase 1)

Service	Monthly Cost (Est.)
Railway/Render (API server)	$20-50
PostgreSQL (managed)	$15-30
Redis (managed)	$10-20
S3/R2 storage (50GB)	$5-10
CDN (Cloudflare)	Free-$20
FCM (Push notifications)	Free

Service	Monthly Cost (Est.)
Domain + SSL	$15/year
Total Phase 1	**~$65-130/month**

Cost at Scale (Phase 3+, 50K users)

Service	Monthly Cost (Est.)
API servers (2-3 instances)	$100-200
PostgreSQL (bigger instance + replica)	$50-100
Redis	$30-50
S3/R2 (500GB + bandwidth)	$30-50
CDN	$20-50
Search (Meilisearch)	$30
Monitoring (Sentry, logs)	$30-50
Total at Scale	**~$290-530/month**

Risk Assessment

Risk	Impact	Probability	Mitigation
Low user adoption	High	Medium	Focus on UX, gamification, local communities
Content moderation challenges	Medium	High	AI auto-flag + community reporting + manual review
Scale issues with PostGIS queries	Medium	Low	Proper indexing, caching, pagination
Photo storage costs	Medium	Medium	Image compression, lifecycle policies, CDN caching
Real-time chat scaling	Low	Medium	Redis pub/sub, WebSocket clustering
API abuse / spam reports	Medium	Medium	Rate limiting, captcha, user reputation system
Data privacy concerns	High	Low	GDPR compliance, encryption, clear privacy policy